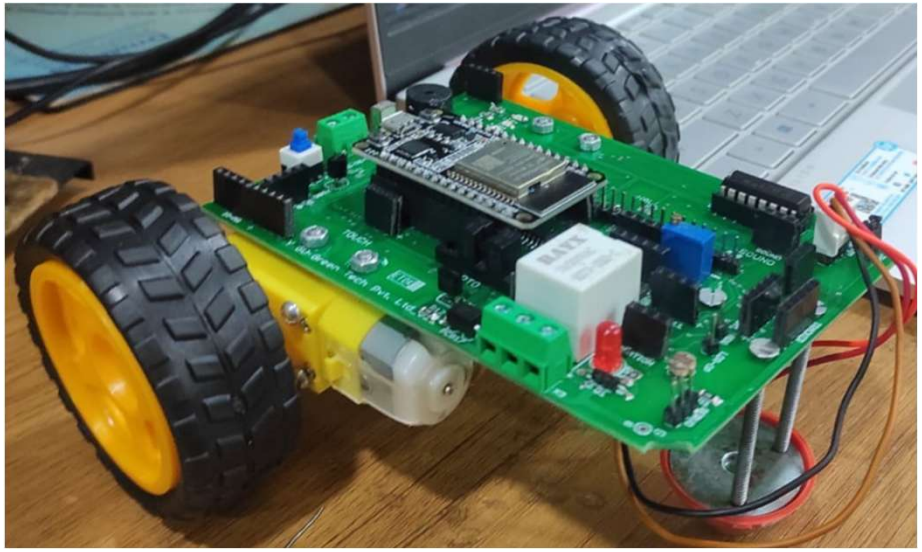
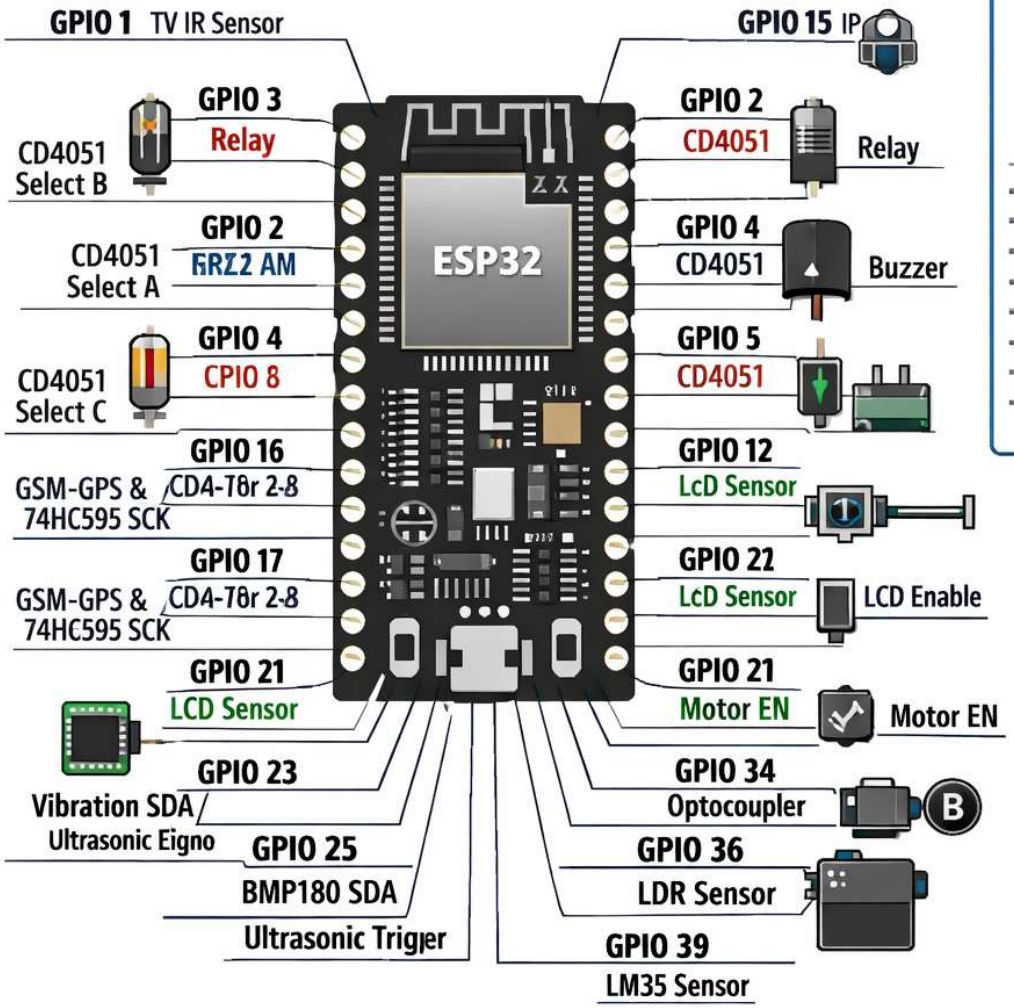
Internet of Things (IoT)
EDU_ROBO IoT Trainer Kit

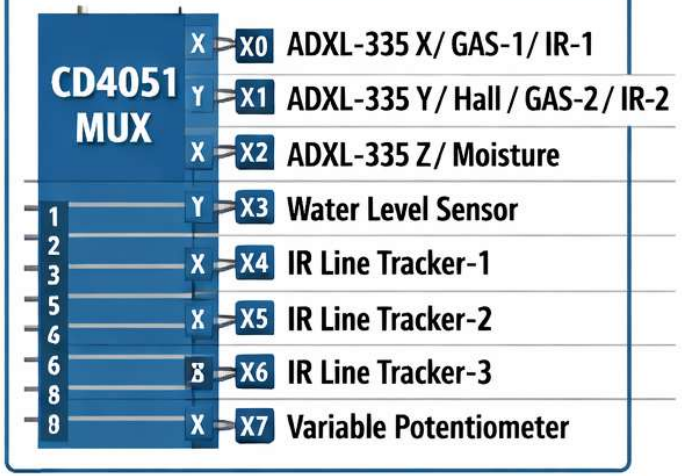
HARDWARE Laboratory Manual-V4

Yerramreddy's
Go-Green Technologies Pvt. Ltd.
GSTN: 27AABCY4791J1ZS
SP-TBI, 8th Floor, Sardar Patel Institute of Technology, Munshi Nagar, Bhavans Campus, Andheri (West), Mumbai-58, India
Web: yrkmgstc.in E-Mail: gogreenramakrishna@gmail.com Mob: 9820962870/9137692917





CD4051 Analog Mux Channels



CD4051 Control Lines

Control Pin	ESP32 GPIO
Select Line A	GPIO 4
Select Line B	GPIO 2
Select Line C	GPIO 5

ESP-32 GPIO Pin Mapping (Ordered)

GPIO Pin	Connected Device / Function
GPIO-1	TV IR Sensor (TSOP1838)
GPIO-2	CD4051 Select Line B
GPIO-3	Relay
GPIO-4	CD4051 Select Line A
GPIO-5	CD4051 Select Line C
GPIO-12	LED-3 & L293D Motor Driver 3A
GPIO-13	LED-4 & L293D Motor Driver 4A
GPIO-14	LED-2 & L293D Motor Driver 2A
GPIO-15	Buzzer
GPIO-16	GSM-GPS Module & 74HC595 SCK
GPIO-17	GSM-GPS Module & 74HC595 SER
GPIO-18	Touch Sensor
GPIO-19	DHT-11 Sensor
GPIO-21	74HC595 RCK & LCD Enable
GPIO-22	L293D Motor Driver EN
GPIO-23	Vibration Sensor
GPIO-25	BMP180 SDA & Ultrasonic Trigger
GPIO-26	BMP180 SCL & Ultrasonic Echo
GPIO-27	LED-1 & L293D Motor Driver 1A
GPIO-32	Sound Sensor
GPIO-33	Servo Motor
GPIO-34	Optocoupler
GPIO-35	LDR Sensor
GPIO-36	CD4051 Multiplexer Output
GPIO-39	LM35 Temperature Sensor

CD4051 Analog Multiplexer Channel Assignment

CD4051 Channel	Connected Sensor
X0	ADXL-335 (X-Axis) / GAS-1 or IR-1
X1	ADXL-335 (Y-Axis) / Hall Sensor / GAS-2 or IR-2
X2	ADXL-335 (Z-Axis) / Moisture Sensor
X3	Water Level Sensor
X4	IR Line Tracker-1
X5	IR Line Tracker-2
X6	IR Line Tracker-3
X7	Variable Potentiometer

CD4051 Control Lines Summary

CD4051 Pin	ESP-32 GPIO
Select Line A	GPIO-4
Select Line B	GPIO-2
Select Line C	GPIO-5

Salient Features

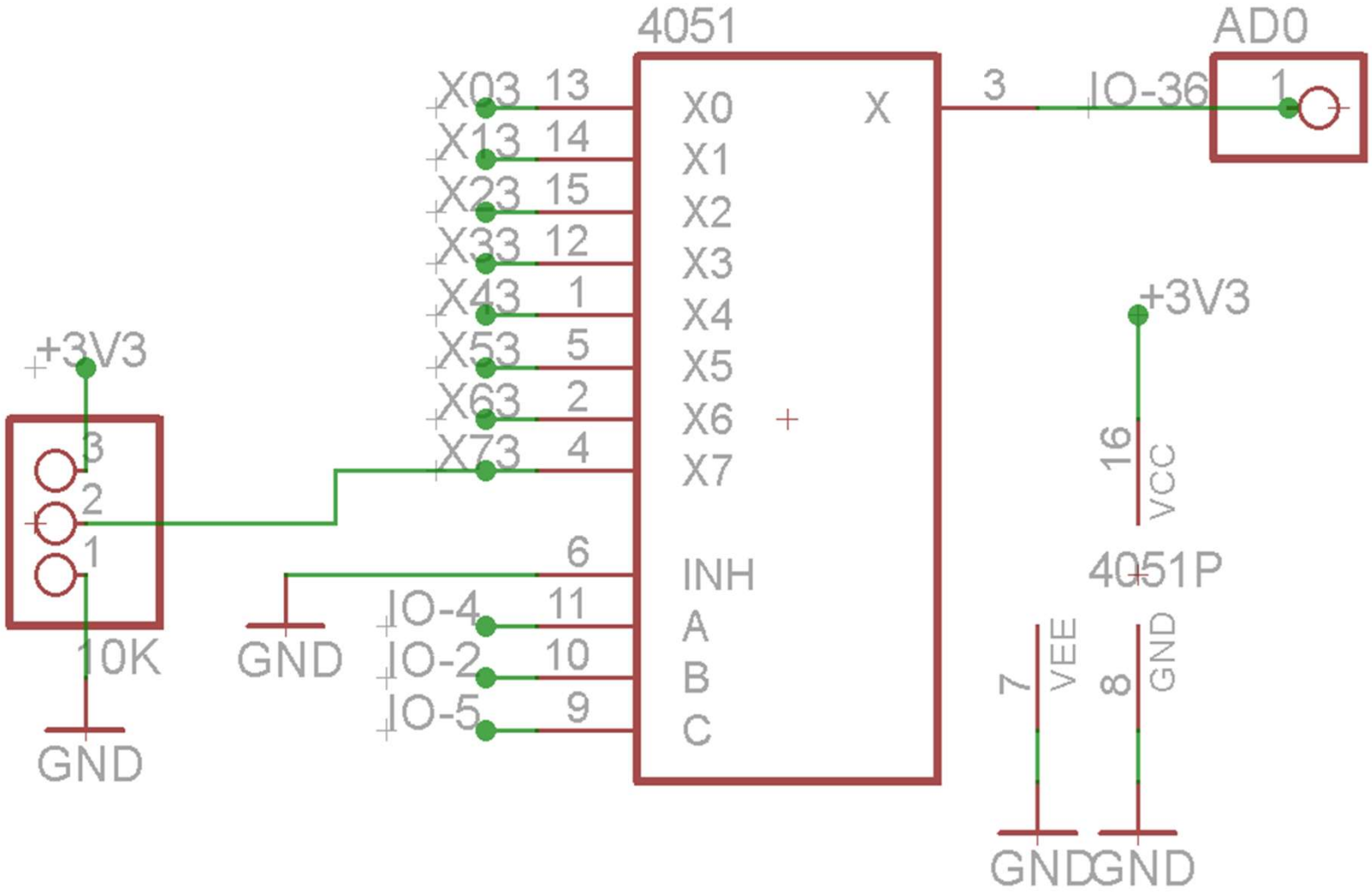
- ESP-32 RISC controller-based Data Acquisition System
- DC Supply of +12 V, +5V, +3.3V
- Provision for piggyback mounting of other CPUs through Female Burg Connector
- Input & Output Interface: Buzzer, 4-LEDs
- Sensors: LDR, DHT-11, LM35, HALL, Optocoupler, DHT-11,Ultrasonic, TV IR
- L293D Motors Interface, Relay, Actuator, STEPPER Motor, DC Motor, and DC Servo
- LCD 16x2 Display through Serial to Parallel Converter 74HC595
- IR Sensors through Analog Multiplexer CD4051
- Wi-Fi/ Bluetooth-based Computer/Mobile/Cloud interfacing and Data Acquisitions and LMS
- Zero PCB, Male and female connectors

Hardware Operational Guidelines

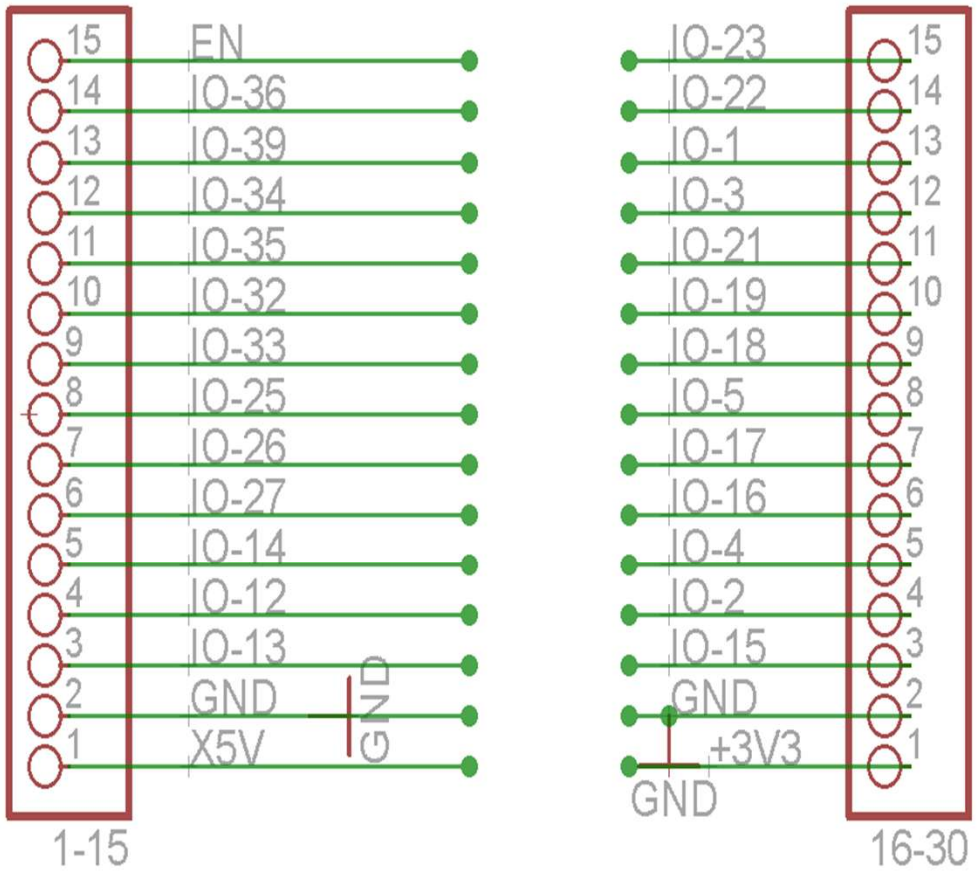
- Switch OFF the DC Supply when mounting any external connecting wires.
- After completion of the wiring, ensure no short circuits and reverse connection of Supply and Ground pins. Any wrongdoing damages the power supply and permanently damages the electronic components.
- Most of the components have a direction. Verify which is the PIN number one in the case of ICs. In the case of LEDs, and Diodes find out the Anode and Cathode terminals. In the case of external sensors, the Supply and ground pins.
- Verify the rating of the components before interfacing i.e., Voltage and Current rating. Any kind of excess loading and reverse polarity may damage the electronic components.
- When giving external Voltage to Digital ICs, and microcontrollers make sure no signal voltage exceeds 3.3V (i.e., 0 to 3.3V, no negative signal) and do not draw a current of more than 10 mA from IC pins.
- While writing the programs for LCD, Stepper Motor, DC Motor, Servo Motor, and Relay refer to the Connector details of the manual to know where these signals are available for interfacing purposes.
- When you switch ON the power to the board using Push to ON switch the +5V LED indicator shows power is OK. Similarly, ESP-32 uses the external USB-Micro B type cable for programming and giving power to the CPU, LCD, and DDS-Function Generator. The power supply for these components is isolated from the DC supply source of the Board. You can use the 2-pin jumper to give the onboard power to ESP-32.
- Piggyback daughter boards and LCD mounting directions are inside the baseboard.
- If any components are hot or getting a burning smell immediately switch off the AC mains and also remove the USB cable.
- In such case disconnect all the external connections you have made and verify once again. If the problem exists, i.e. POWER LED is not getting ON then return the kit to the institute for maintenance purposes.
- While loading the program in ESP-32 press BOOT Switch, to Restart the Program press RESET Switch

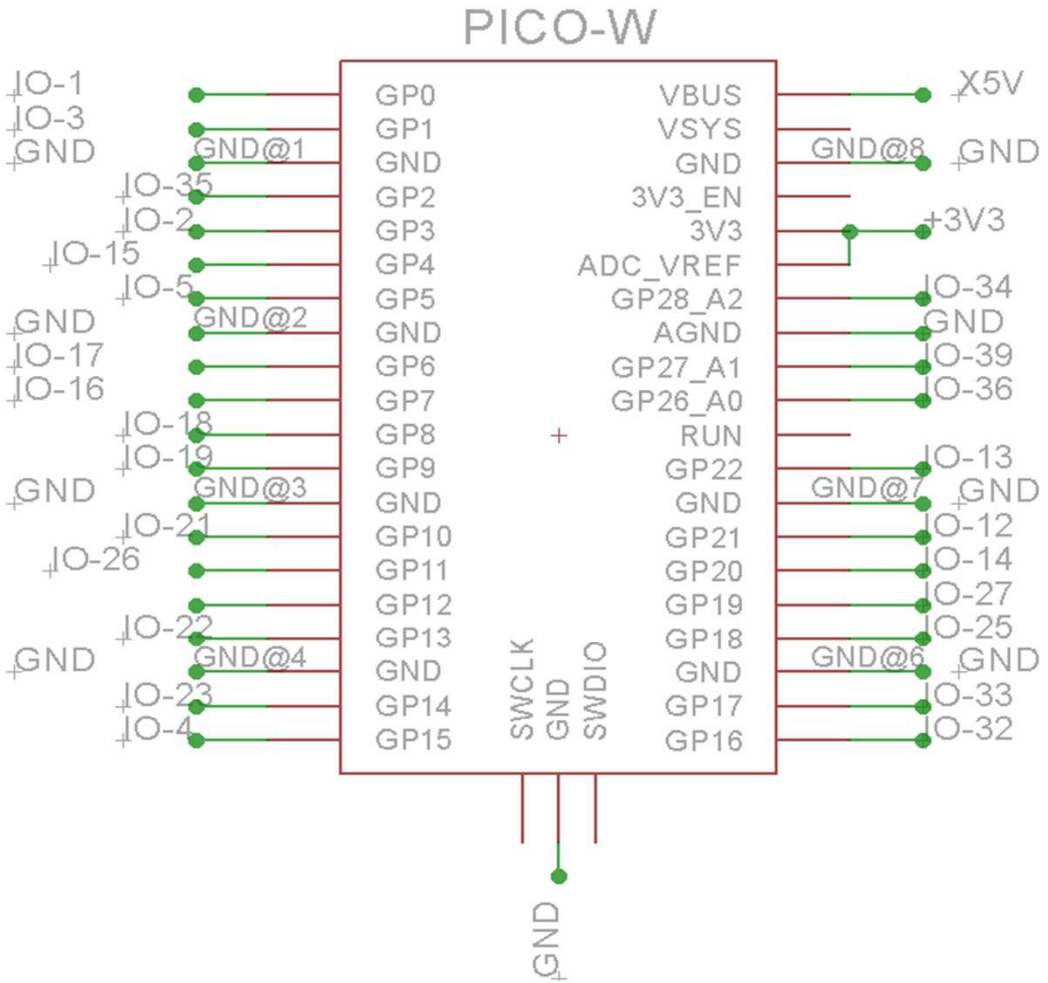
Analog Multiplexer

ESP32: A=IO4, B=IO2, C=IO5



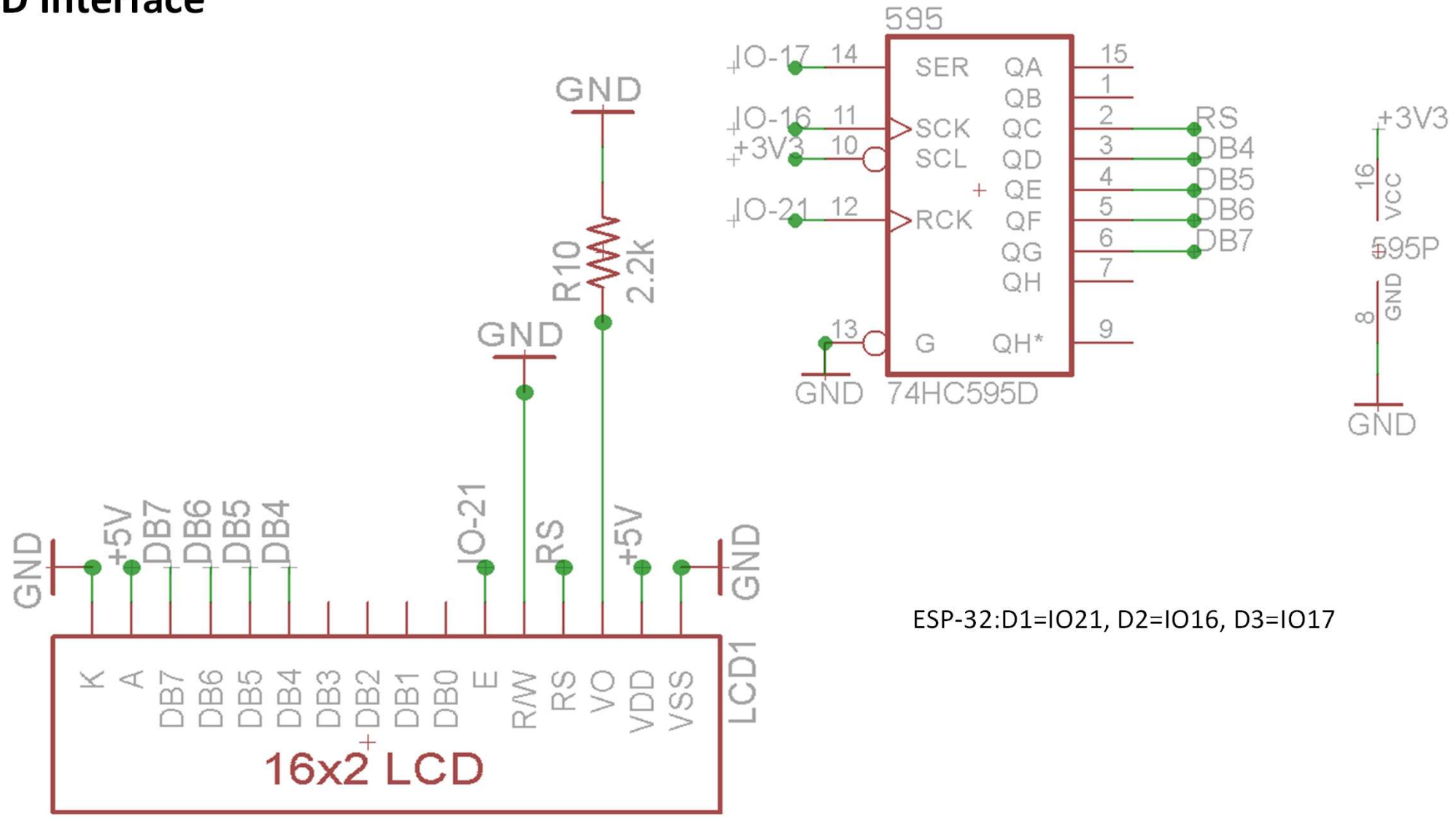
ESP-32 Interface Details



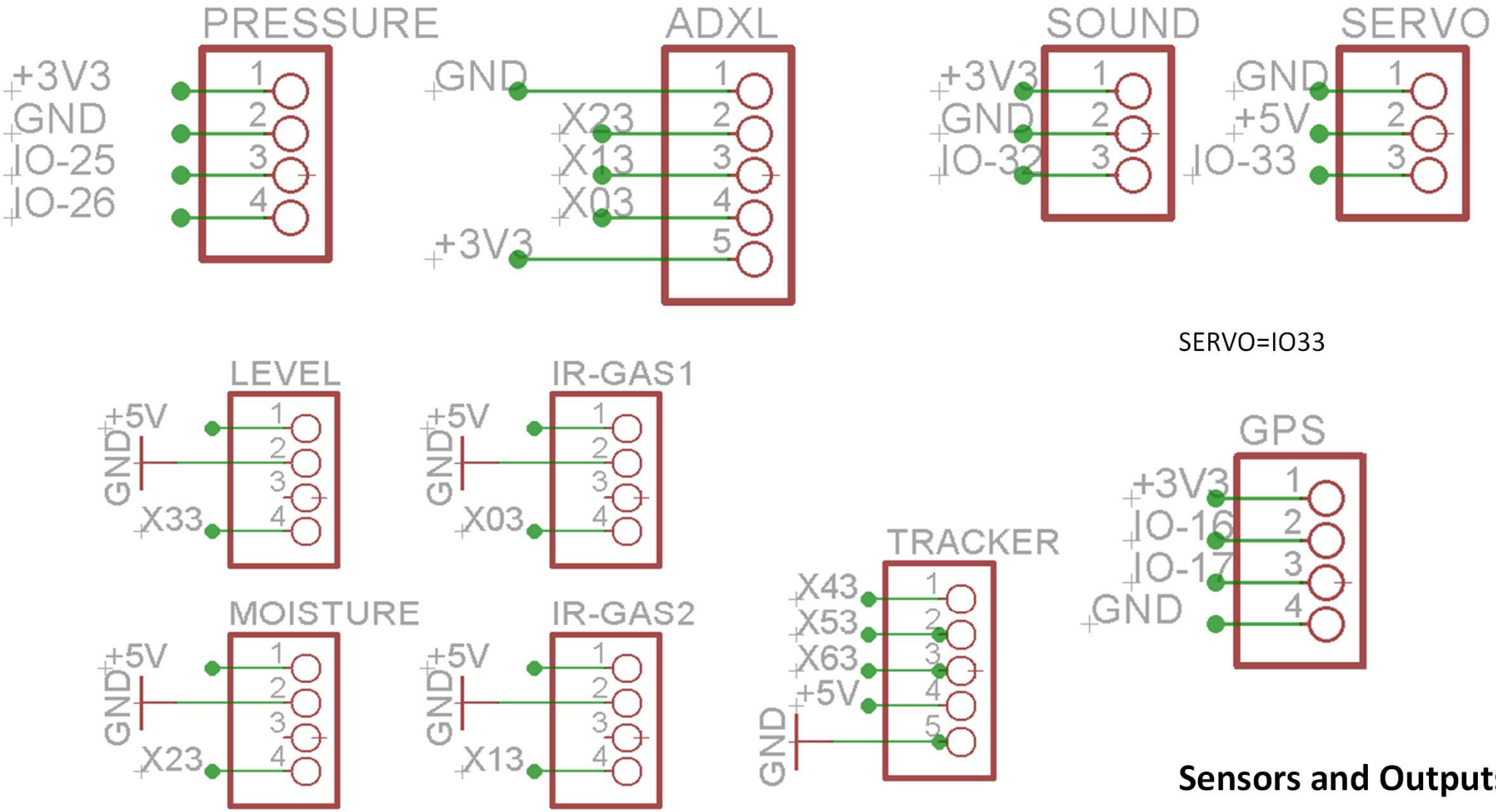


Raspberry Pico-W Interface Details

LCD Interface



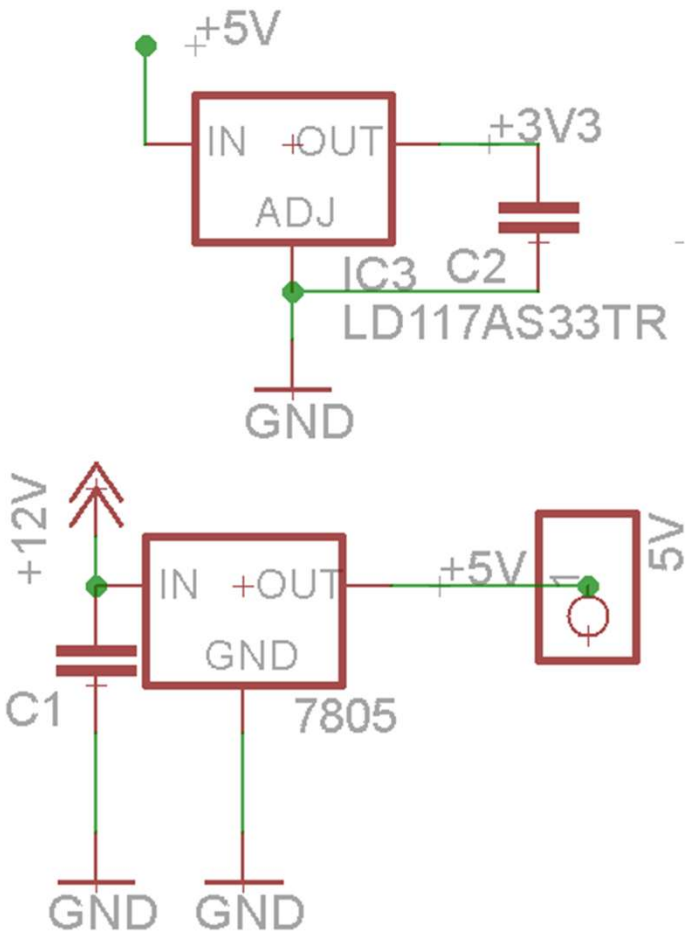
ESP-32:D1=IO21, D2=IO16, D3=IO17

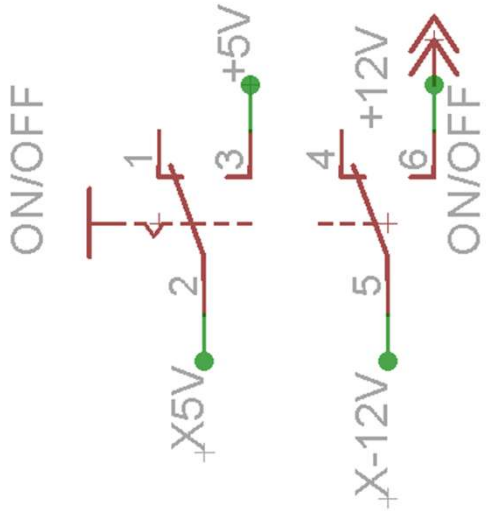


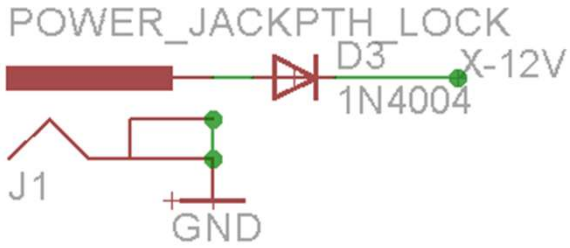
SERVO=IO33

Sensors and Outputs

POWER SUPPLY

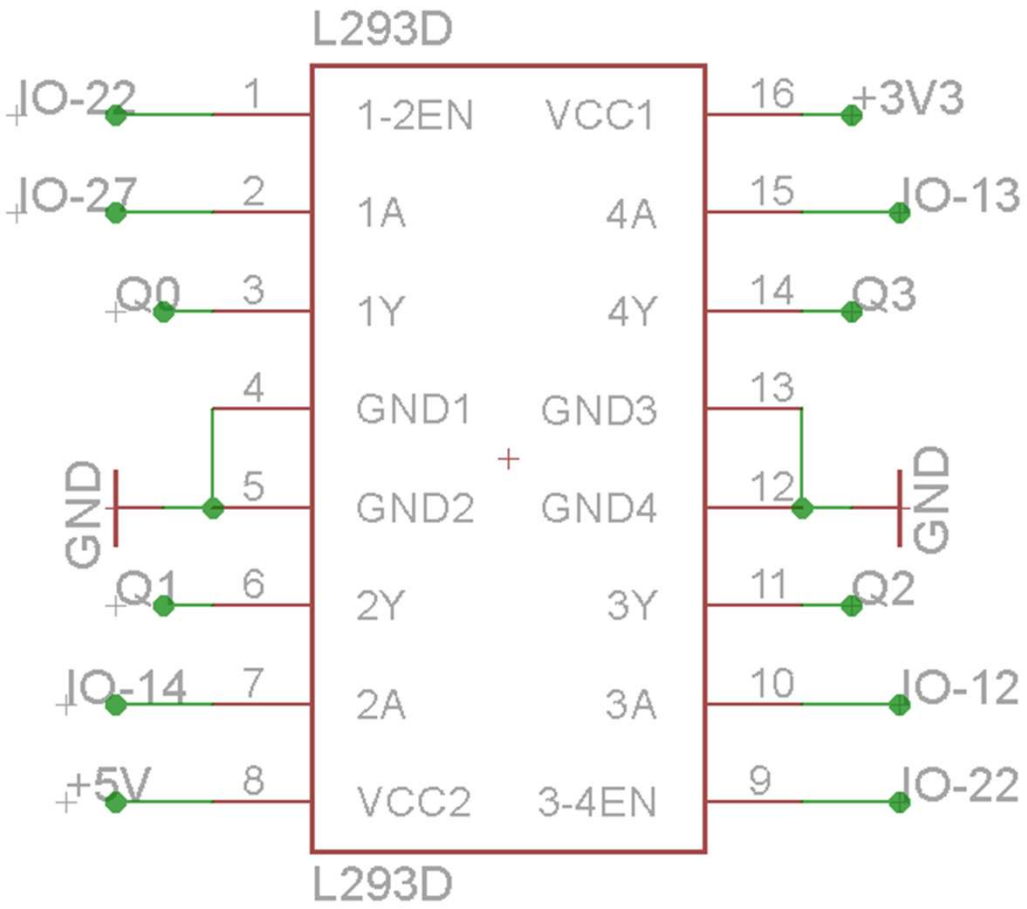


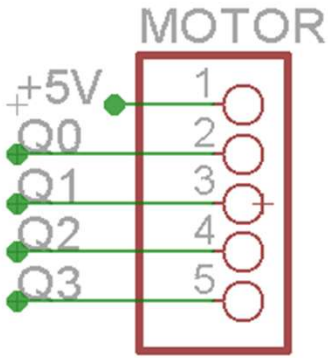




MOTOR DRIVER

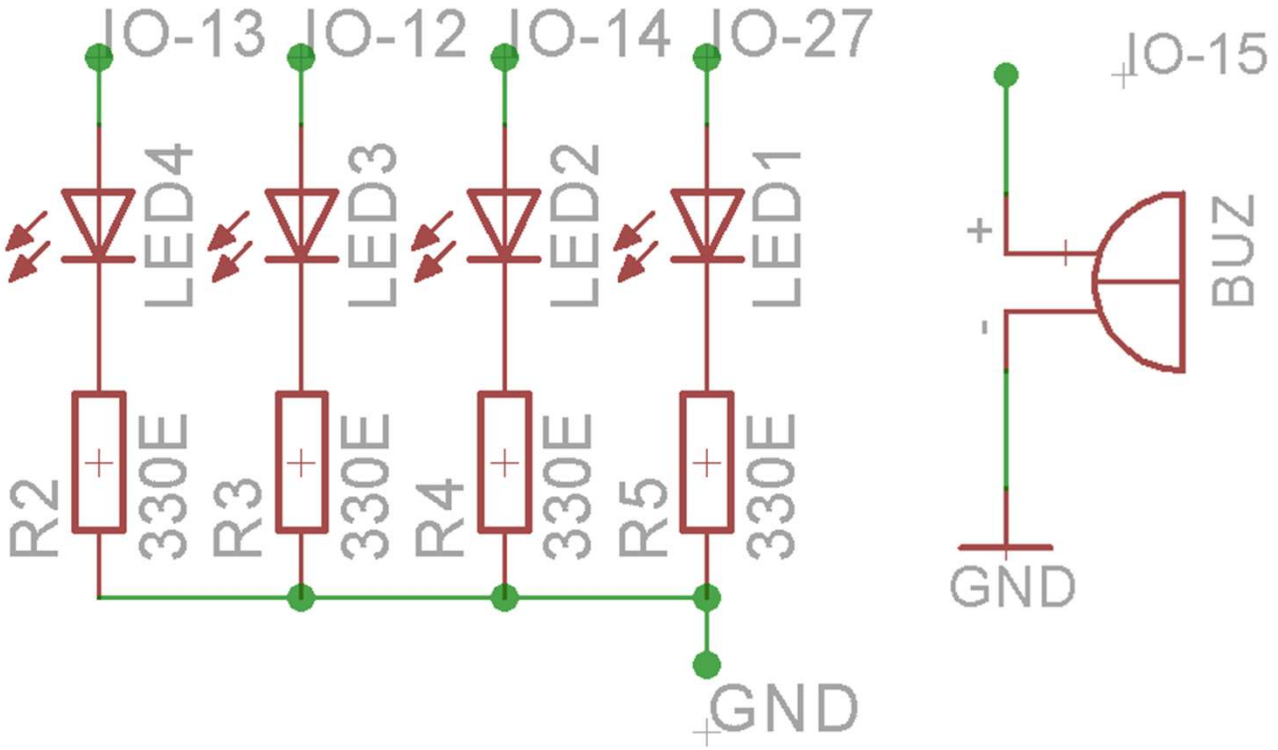
ESP32:
EN=IO22,
1A=IO27,
2A=IO14,
3A=IO12,
4A=IO13





OUTPUT

ESP32:
LED1=IO27,
LED2=IO14,
LED3=IO12,
LED4=IO13

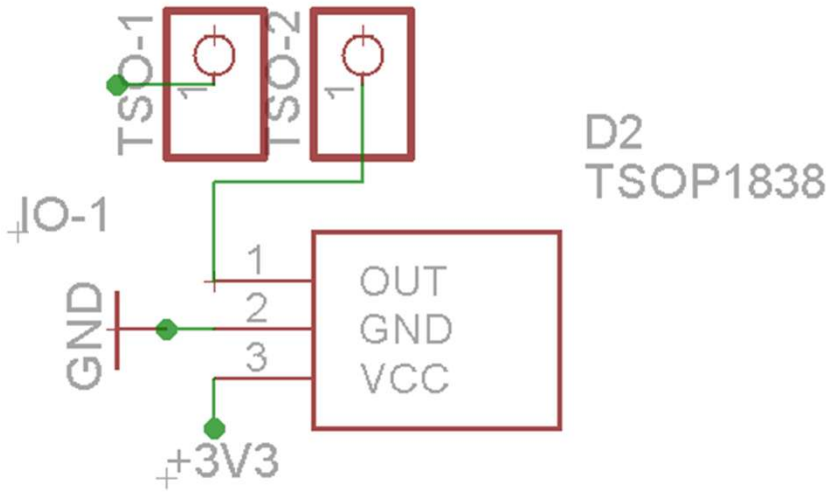


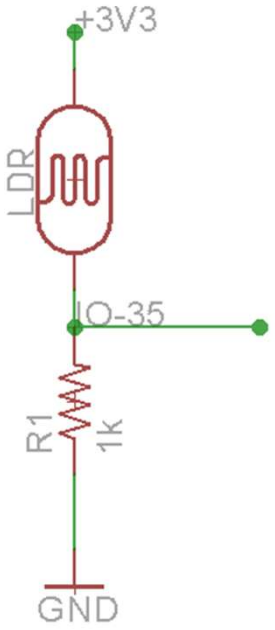
ESP32:
BUZZER=IO15

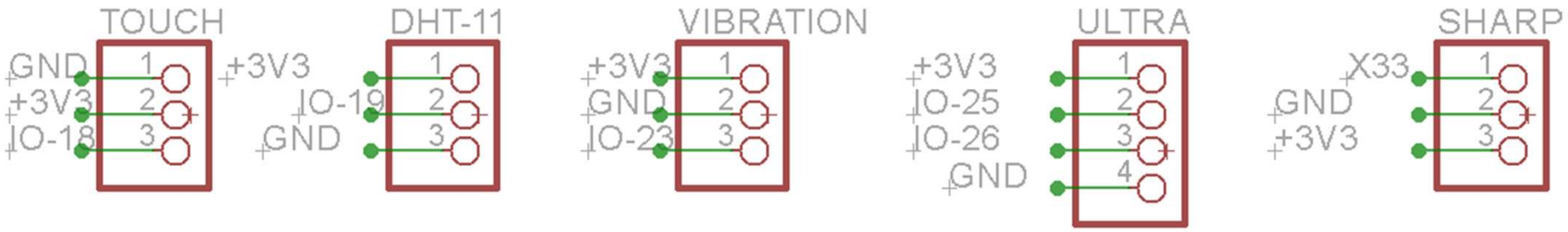
LDR & IR REMOTE, Sensors

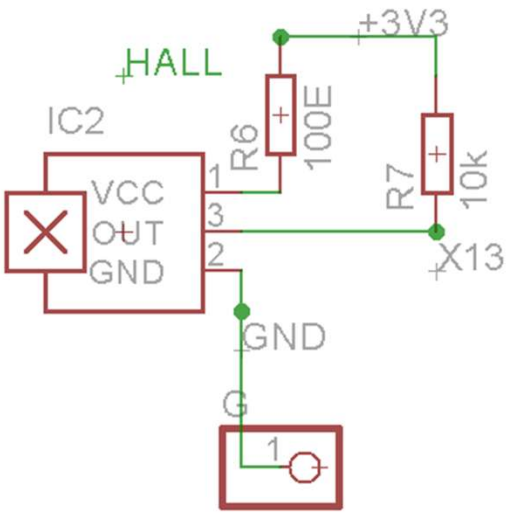
ESP32:LDR=IO35

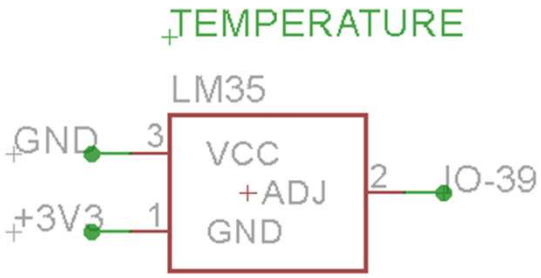
ESP32: TSOP= IO-1

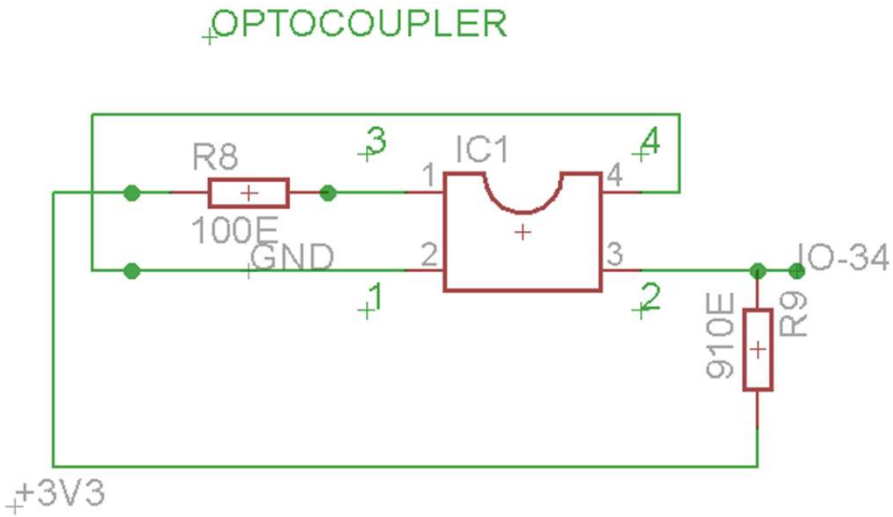












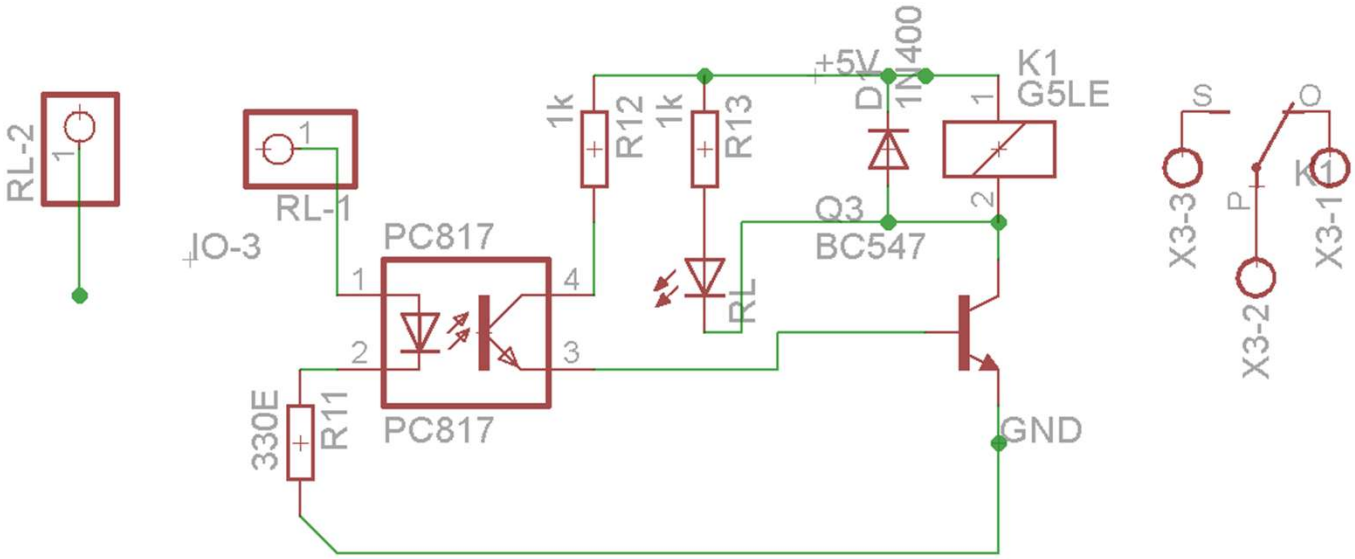
ESP32: HALL=X13

ESP32: LM35=IO39

ESP32: OPTO=IO34

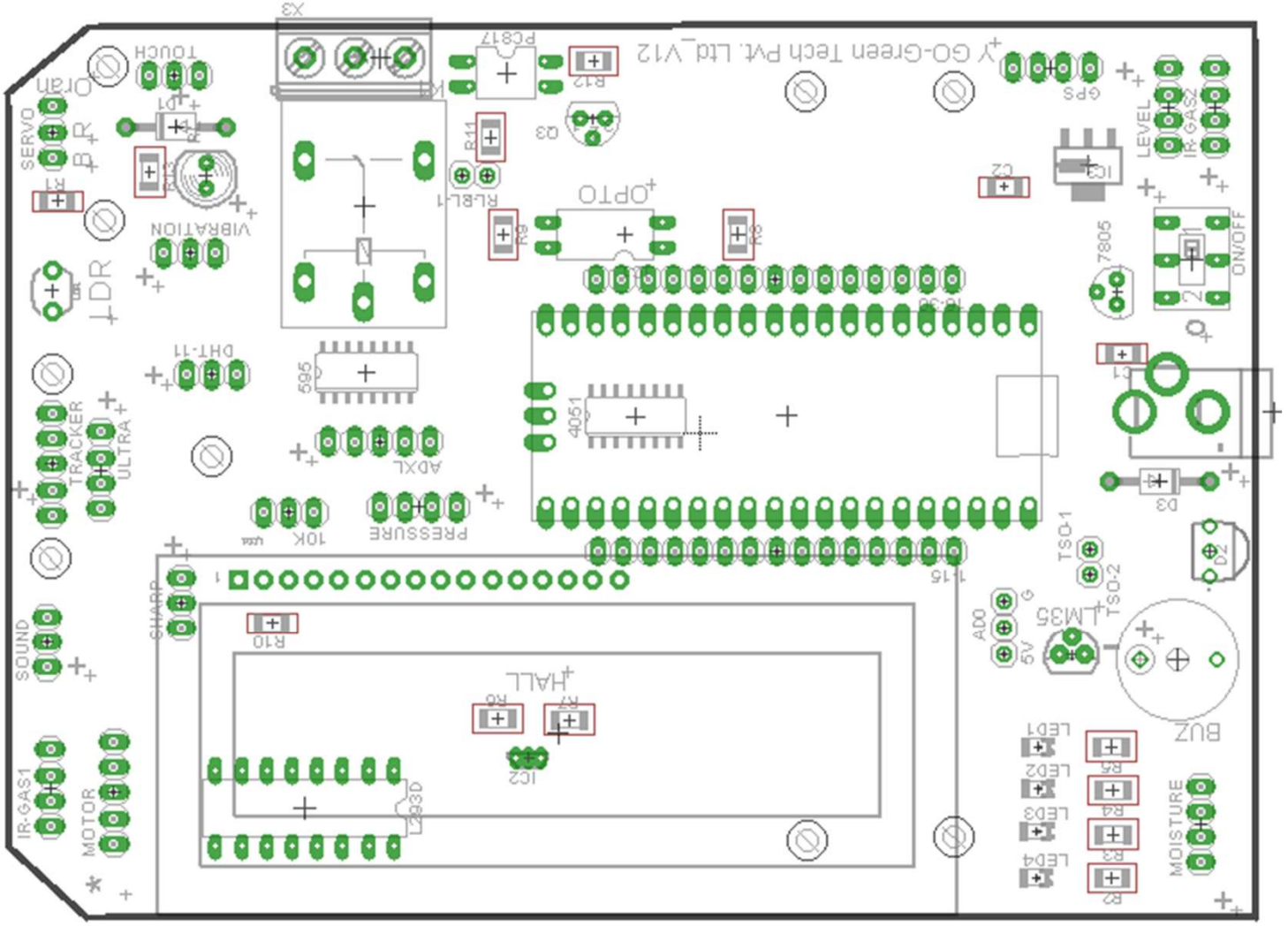
SENSORS Interface

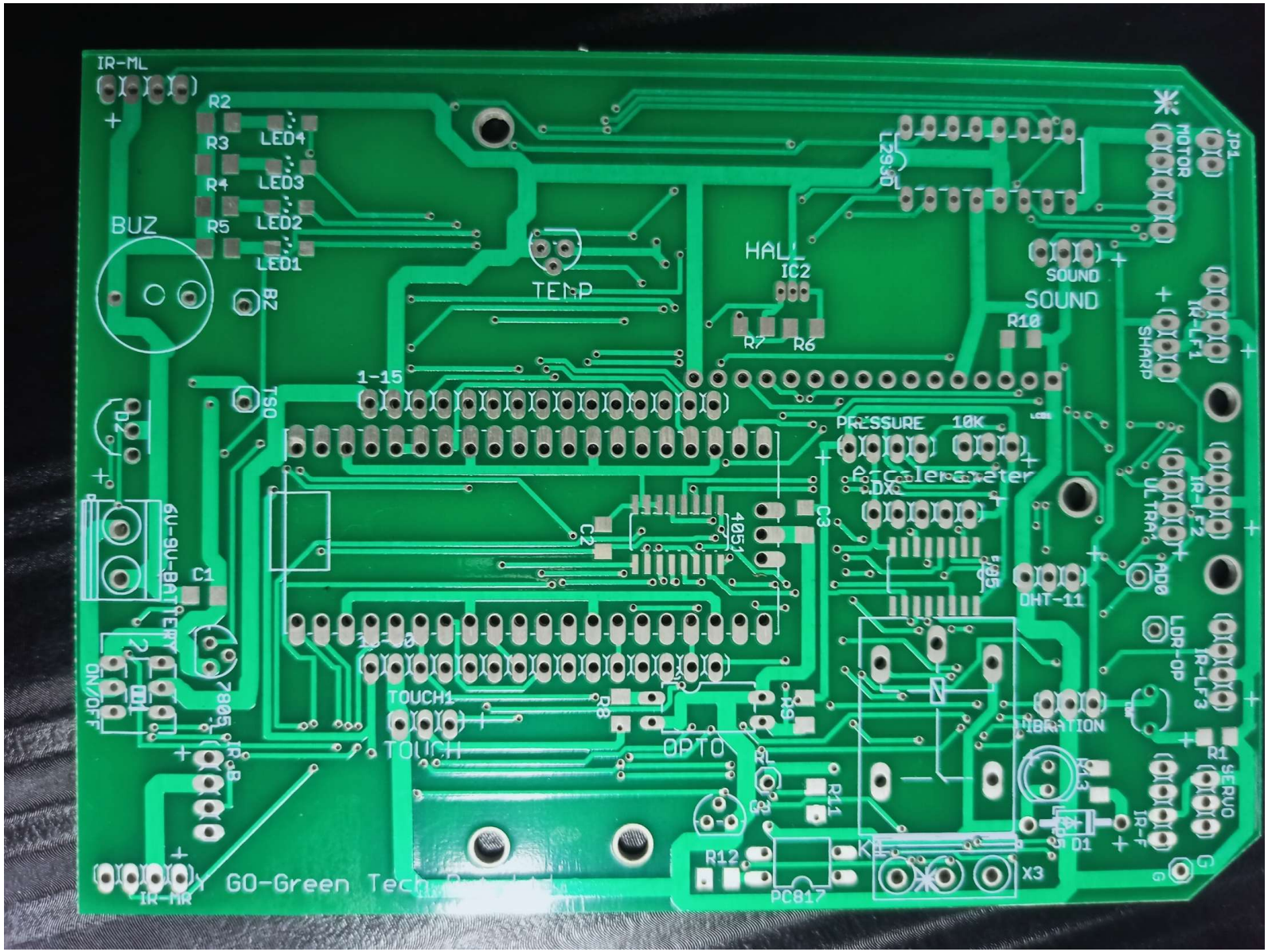
RELAY Interface

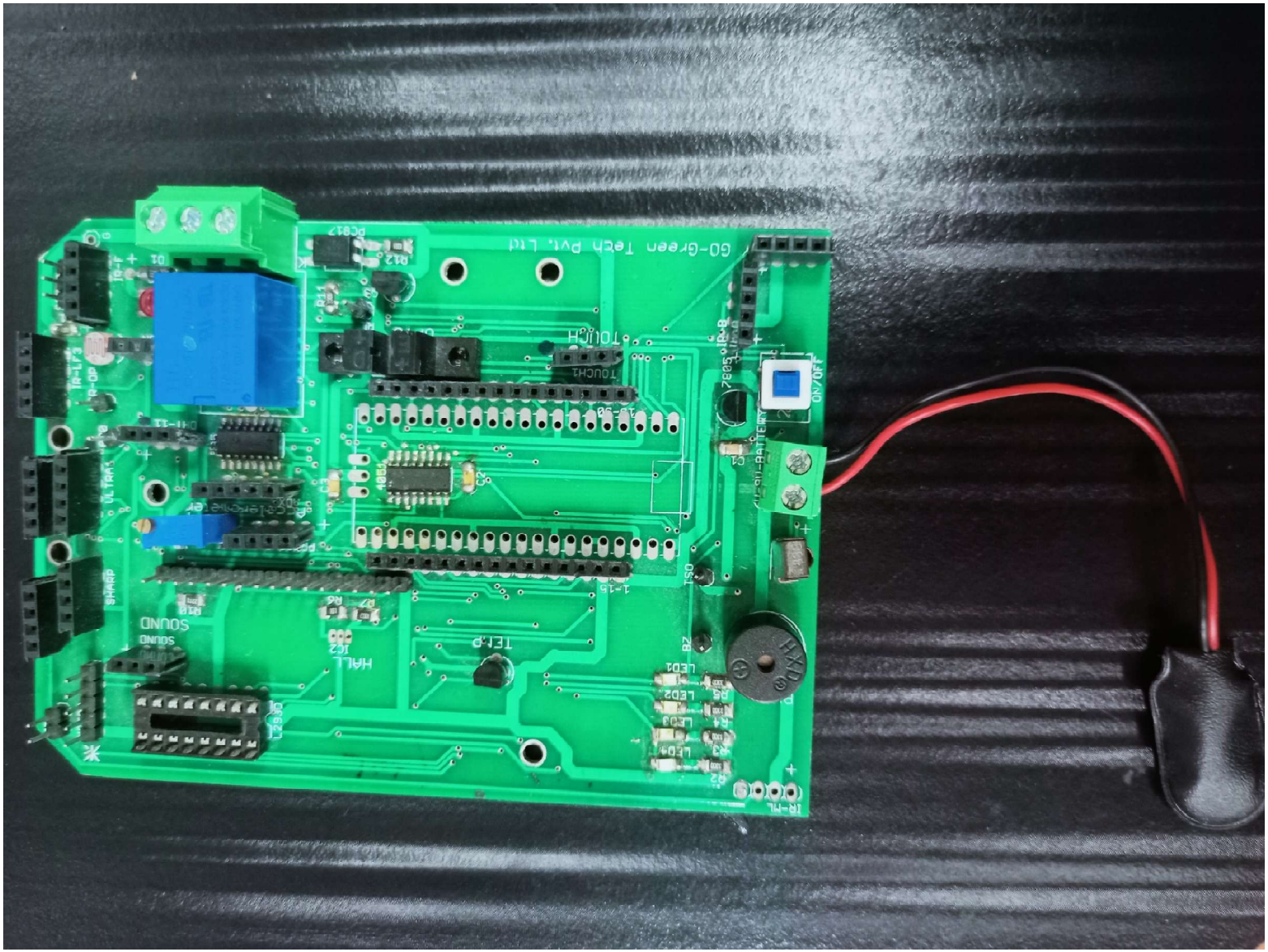


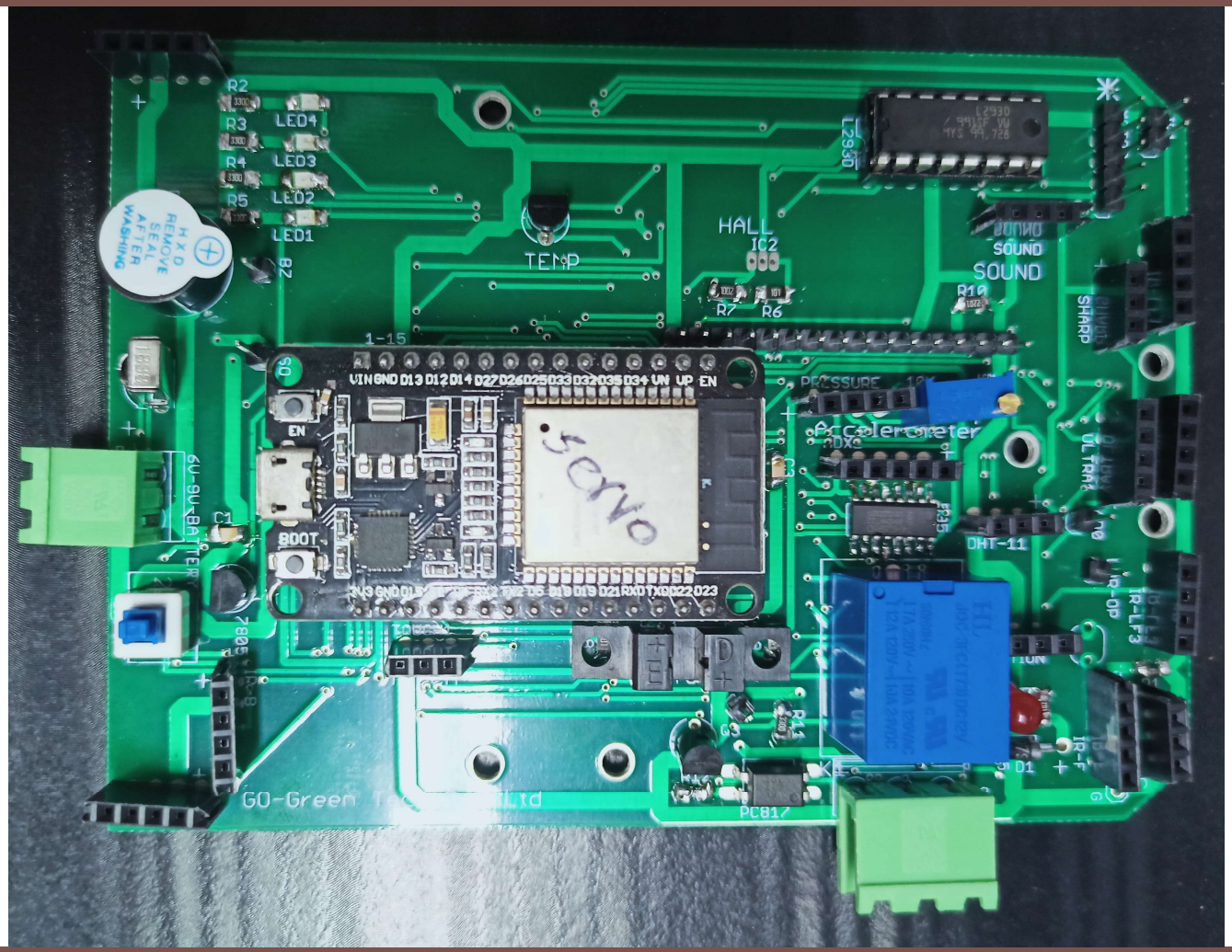
ESP32:
RELAY=IO-3

Component Placement



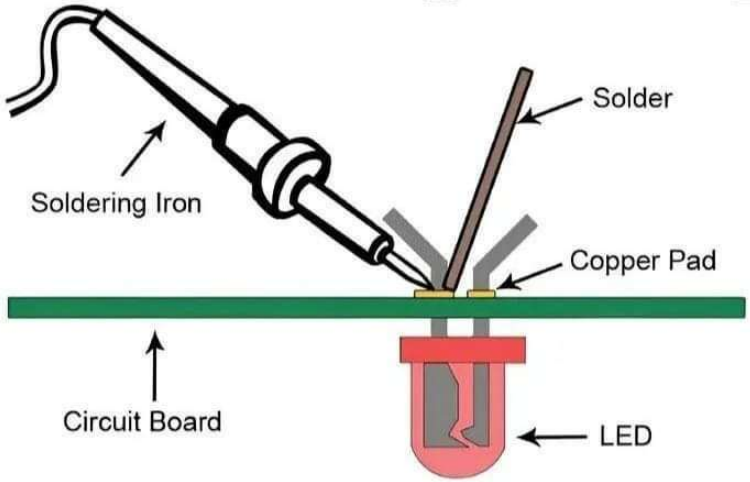

IR-ML
R2
R3
R4
R5
LED4
LED3
LED2
LED1
BUZ
BZ
TEMP
HALL
IC2
L298D
JP1
MOTOR
SOUND
SOUND
R10
R7
R6
SHARP
IR-LF1
1-15
TSO
PRESSURE
10K
DX
IR-LF2
ULTRA
AD0
C3
C2
4051
C1
6V-9V-BATTERY
ON/OFF
7805
DHT-11
LDR-OP
IR-LF3
TOUCH1
TOUCH
R8
R9
OPTO
VIBRATION
R13
D1
IR-F
SERVO
IR-R
R11
R12
PC817
K1
X3
GO-Green Tech

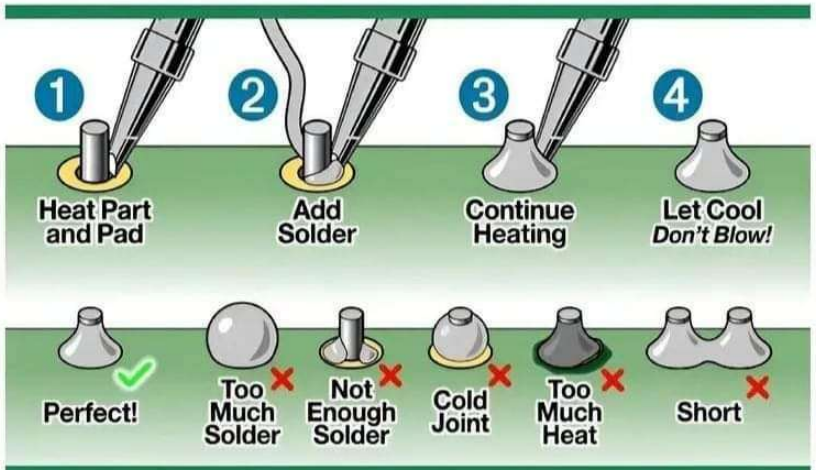

Go-Green Tech Pvt. Ltd
TOUCH
SOUND
HALL
TEMP
ON/OFF


R2
R3
R4
R5
LED4
LED3
LED2
LED1
BZ
TEMP
HALL
IC2
L2930
SOUND
R10
R7
R6
1-15
VIN GND D13 D12 D14 D27 D26 D25 D33 D32 D35 D34 VN VP EN
EN
BOOT
3V3 GND D15 D2 D4 RX2 TX2 D5 D18 D19 D21 RXD TXD D22 D23
Servo
PRESSURE
Accelerometer
SDX
SHARP
ULTRA
DHT-11
IR-OP
IR-LF3
IR-F
6V-9V-BATTER
C1
R11
D1
PC817
GO-Green

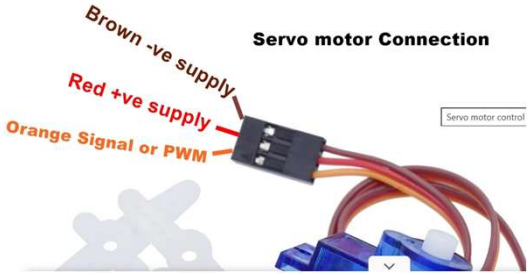
Sr No	Name of the Component	Value
1	R1, R9, R12, R13	1K
2	R2, R3, R4, R5, R11	330 Ohms
3	R6, R8	100 Ohms
4	R7	10 K
5	R10	2.2 K
6	C1,C2,C4,C5,C6	0.1 uF
7	D1 Diode	1N 4004
8	Q3 NPN Transistor	BC 547
9	7805 5V Regulator	7805 TO-92
10	IC	74HC595-S016, CD4051-So-16, L293D, PC817
11	LDR	LDR 3MM
12	TV IR Receiver	TSOP1838
13	BUZZER	BUZZER 5V
14	Ultrasonic Sensor	HC-SR04
15	Optocoupler	MOC7811
16	LED	5MM LED
17	DC Motor 5V DC	BO1 VJT389 5V DC Motor Qty-02
18	RELAY 5V DC	5VT73 Relay
19	LCD	16x2 LCD
20	Motor Clamp	VJT303 CLAMP Qty-02
21	Power Connectors	58/2C Vital, 58/3C Vital
22	USB Cable	Micro USB B-type Cable
23	9V Battery Connector	Black Snap Battery Connector
24	5PIN Motor Connector	SHS-05 Male & Female
25	Dummy Wheel	Dummy Wheel
26	Vehicle Tyers	BO- Tyers Qty-02
27	Microcontroller (VM842)	ESP32-WROOM-32 Devkit V1 (30 PIN)
28	Potentiometer	10k 3296
29	Temperature & Humidity Sensor	DHT-11
30	Push to On Switch	PCB Mount PUSH to ON/OFF Switch
31	Burg Strip 40x1	Male and Female
32	SBC PCB	Go-Green Double-Sided PCB
33	Plastic Box	Plastic Box (1680ML)
34	Connecting wires F-F, F-M	Male and Female Connecting Wires

Soldering Steps

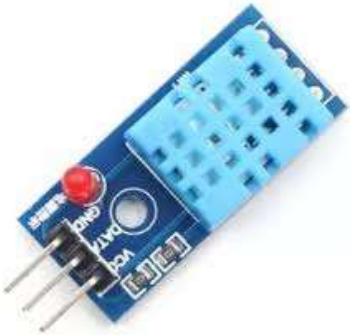




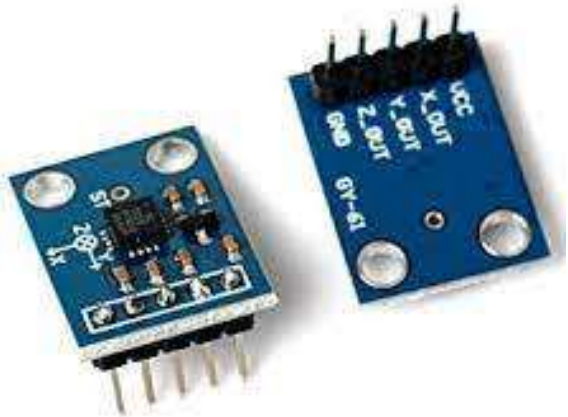
On Board Device Connection Examples-1



SERVO MOTOR: IO33



SERVO MOTOR: IO19



ADXL: Z:X0, Y:X1, X:X2



RELAY: IO3



LDR: IO35



BUZZER: IO15



Ultrasonic: ECHO-GPIO25, TRIG: GPIO26



OPTO COUPLER:IO34

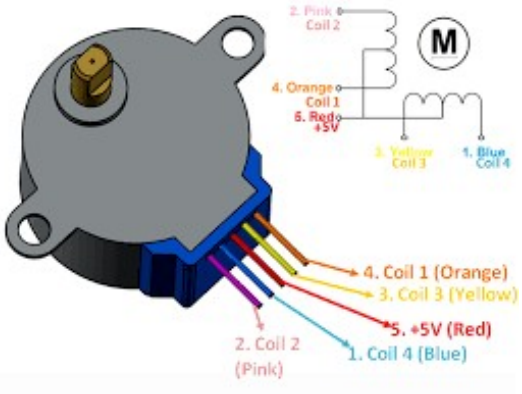


LM35: IO39

On Board Device Connection Examples-2



SOUND: IO32



STEPPERMOTOR:IO27, IO14,IO12,IO13, IO4



TOUCH:IO18



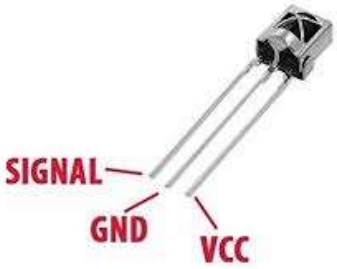
VIBRATION:IO23



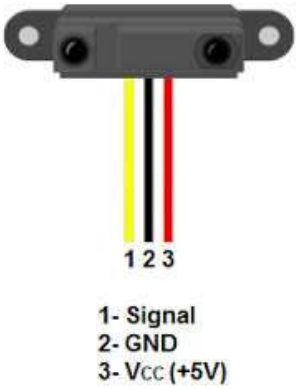
GAS1: X0 GAS2: X1



PRESSURE SENSOR: SCL: IO25 SDA: IO26

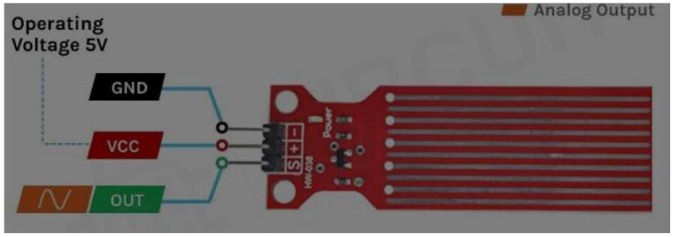


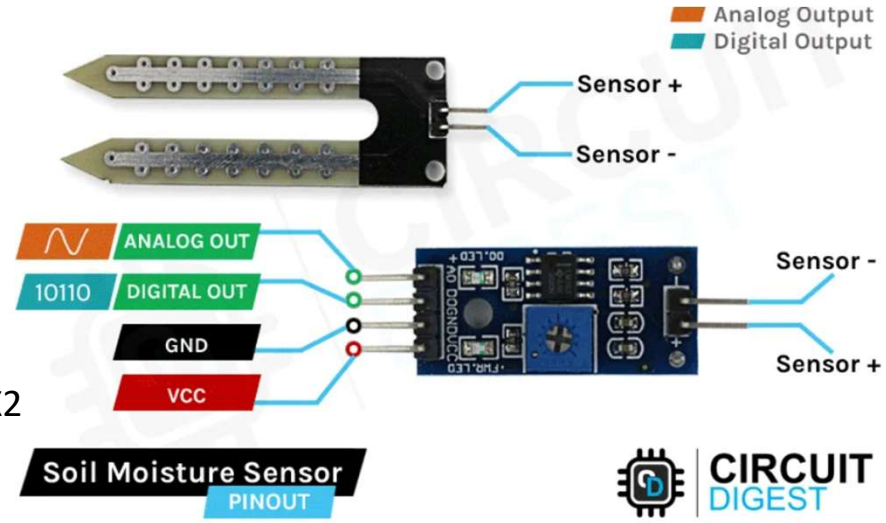
TSOP: IO-1



SHARP:X5

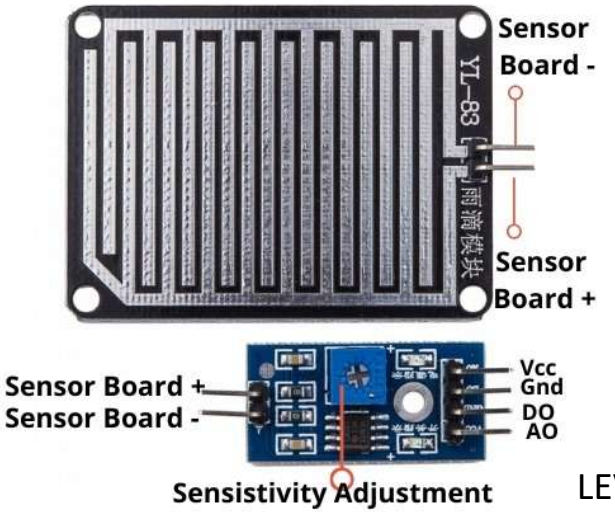
On Board Device Connection Examples-3





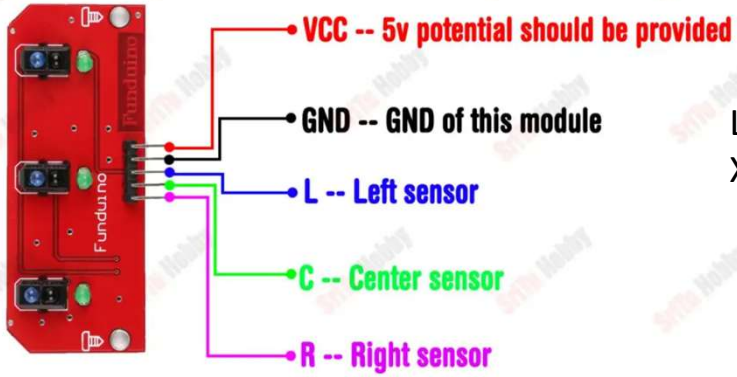
Moisture: X2

DO = Digital Output
AO = Analog Output



LEVEL/RAIN: X3

The PIN structure of this sensor



Line Follower:
X4, X5, X6

Videos

Arduino Compiler:
https://youtu.be/V9SctLdO8_M?si=OTISBMbOxUN3oywT
Arduino Data Types:
https://youtu.be/1ObxuBo77As?si=GxTxJQVa8yBkxEFL
https://youtu.be/hSPkSkTQaGg?si=lpLZNd9t5ekg977d
https://youtu.be/ekNysFMo064?si=keSGqHr-0OrVpr6k
https://youtu.be/FF_MnFxjSGE?si=djyMtt5H9EeOBZN-
https://youtu.be/f9xvhRSAt6U?si=54jSu4ITWf3PRMxO
ESP-32 EDU-ROBO:
Session-1:ESP-32 Hardware.:
- https://youtu.be/_jfXpjCVybQ?si=cnHh3MrcmtA5UEAT

Session-2:Software Configuration:
- https://youtu.be/SV5Wy4jKDRY?si=1xEp-cDUUT8Kaqar

Session-3:Signals and Systems Part-1:
- https://youtu.be/3MaVT3dO2c8?si=2pCDpqFf4AwVC_nO

Session-4:Signals and Systems Part-2
- https://youtu.be/nKiOYB_03qw?si=oWYzdUp5xxX7W8IM

Session-5:LED and Buzzer Interface:
- https://youtu.be/Wquhak4PkR0?si=EkHecjkbWk10ZvyD

Session-6:Input PIN Configuration
- https://youtu.be/qfPNZReEXYQ

Session-7:Soldering SMD Resistors, LEDs and CPU Burg Strip:
- https://youtu.be/InCAWbSKFWQ

Session-8: Soldering of the remaining Componentes:
- https://youtu.be/wNJyIoVtGD8

Session-9: Projects, Products, Research, Publications & Patents Part-1 (Introduction, Big Data Analysis (AI/ML/Deep Learning)
- https://youtu.be/8yhzvvTsrqw

Session-10: Projects, Products, Research, Publications & Patents Part-2 (IoT Introduction, various Levels of Applications)
- https://youtu.be/eQa5PFdoZj0

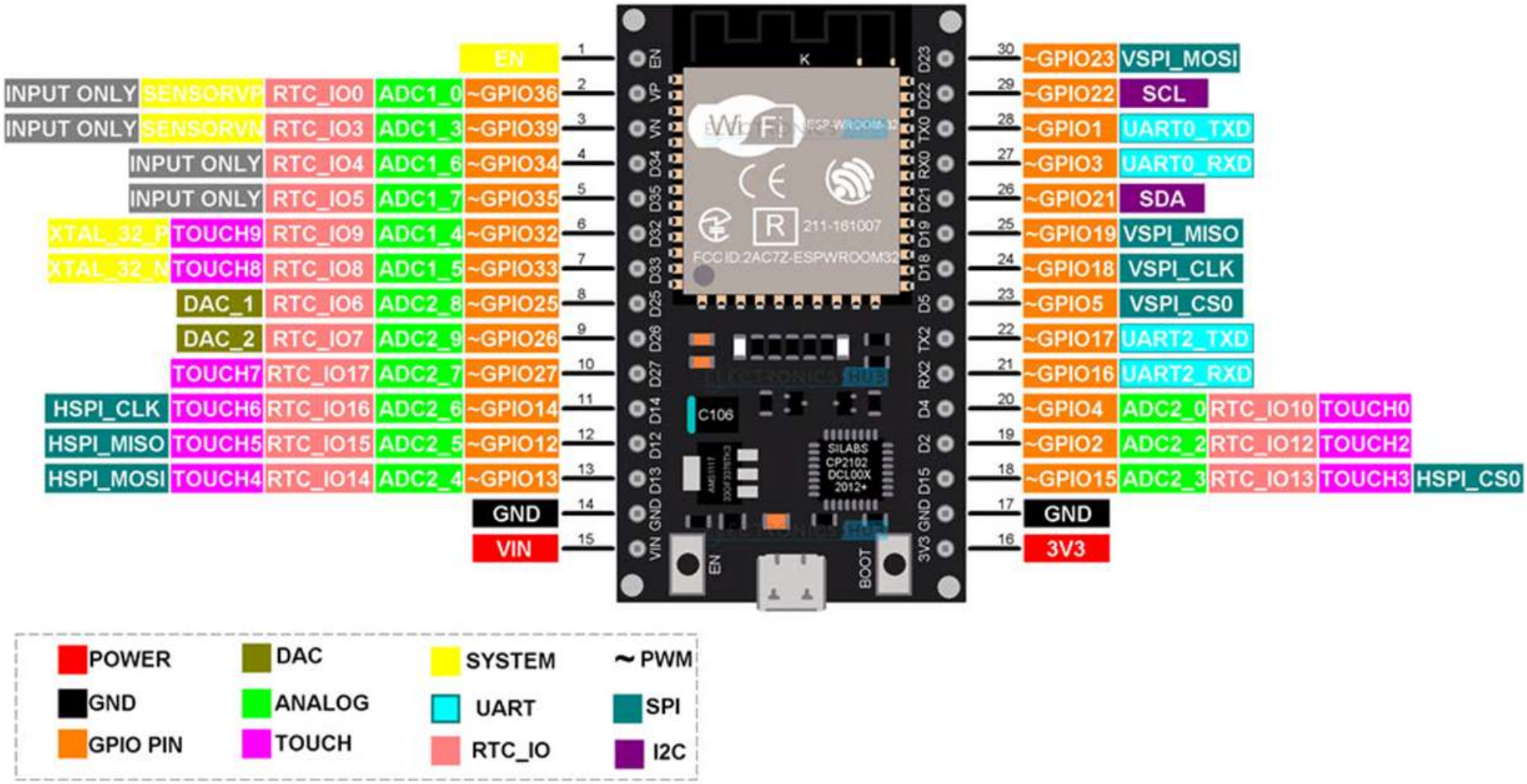
Session-11: Projects, Products, Research, Publications & Patents Part-3 (Scope of IoT Applications and case Studies)
- https://youtu.be/IFJDrkkGXFQ

Session-12: Projects, Products, Research, Publications & Patents Part-4 (Product Proposal Writing)
- https://youtu.be/mrdewNLR9dc

SPI/I2C/CAN/UART Programming:

https://dev.blues.io/blog/blues-university-understanding-sensor-interfaces-uart-i2c-spi-can/

Lecture Videos Link: https://www.youtube.com/user/EMTRONTechnologies



Installation & Options of ARDUINO

Goto Preferences-> enter the following URL at the additional board manager URLs:-

https://dl.espressif.com/dl/package_esp32_index.json

esp32 by **Espressif Systems**
Boards included in this package:
ESP32 Dev Module, WEMOS LoLin32, WEMOS D1 MINI ESP32.
More info

Install ESP32

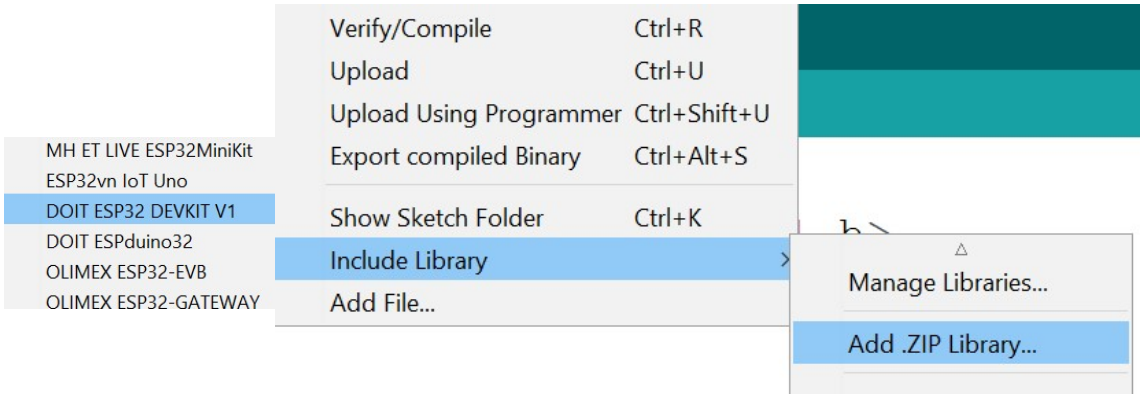
Board Name: DOIT ESP32DEVKIT-V1

Program: Press & Release BOOT key while downloading the program

Sketch-> Include library -> Manage library -> MD_AD9833

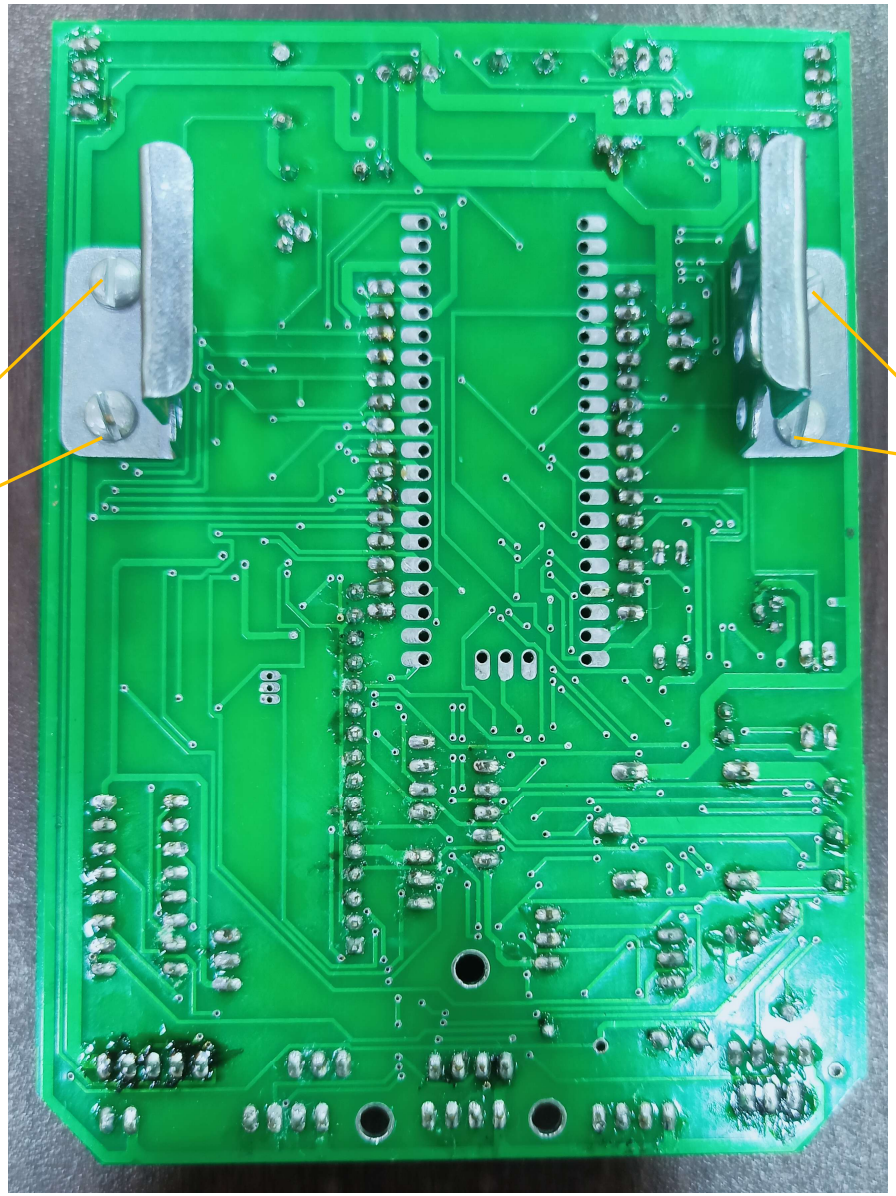
Install:- Blynk, IRremote, DHT_sensor_library, LiquidCrystal-master,

ServoESP32, Stepper, ThingSpeak, Tone32-master, EduIntro





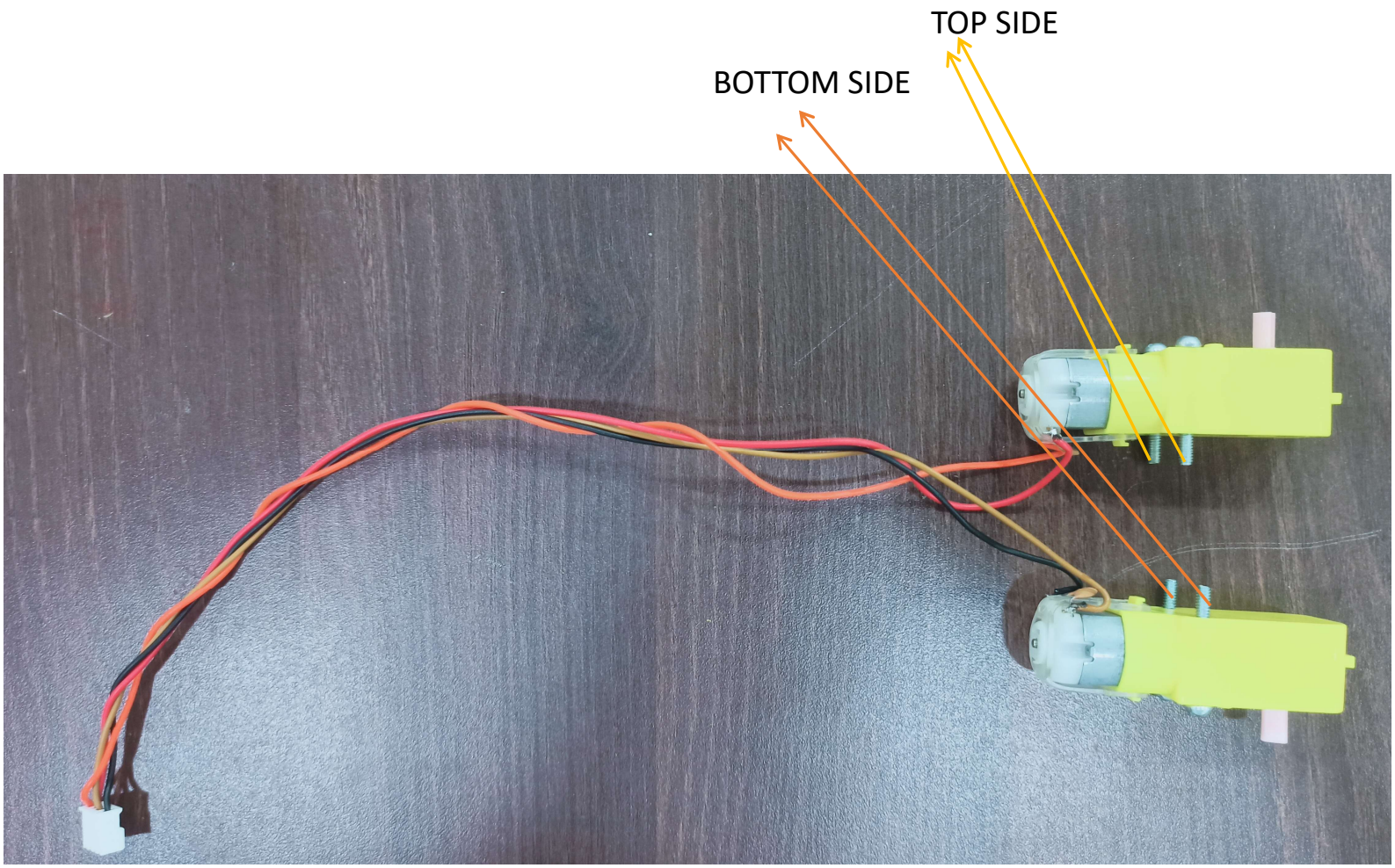
Step-1:
Fix the Motor Clams with 10 mm Screws from the Bottom Side of the PCB

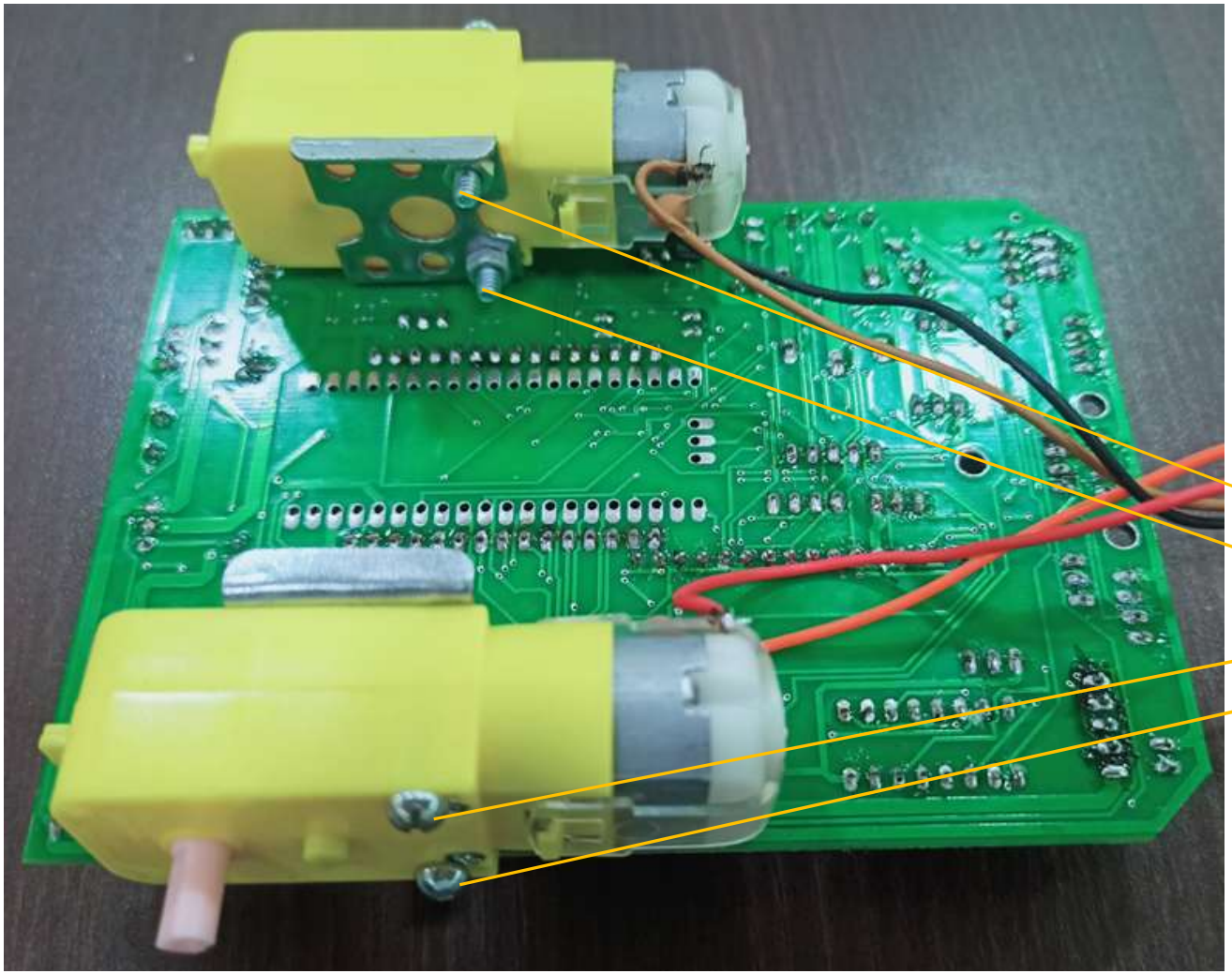


VJT303 MOTOR CLAMP & 10 mm Screws

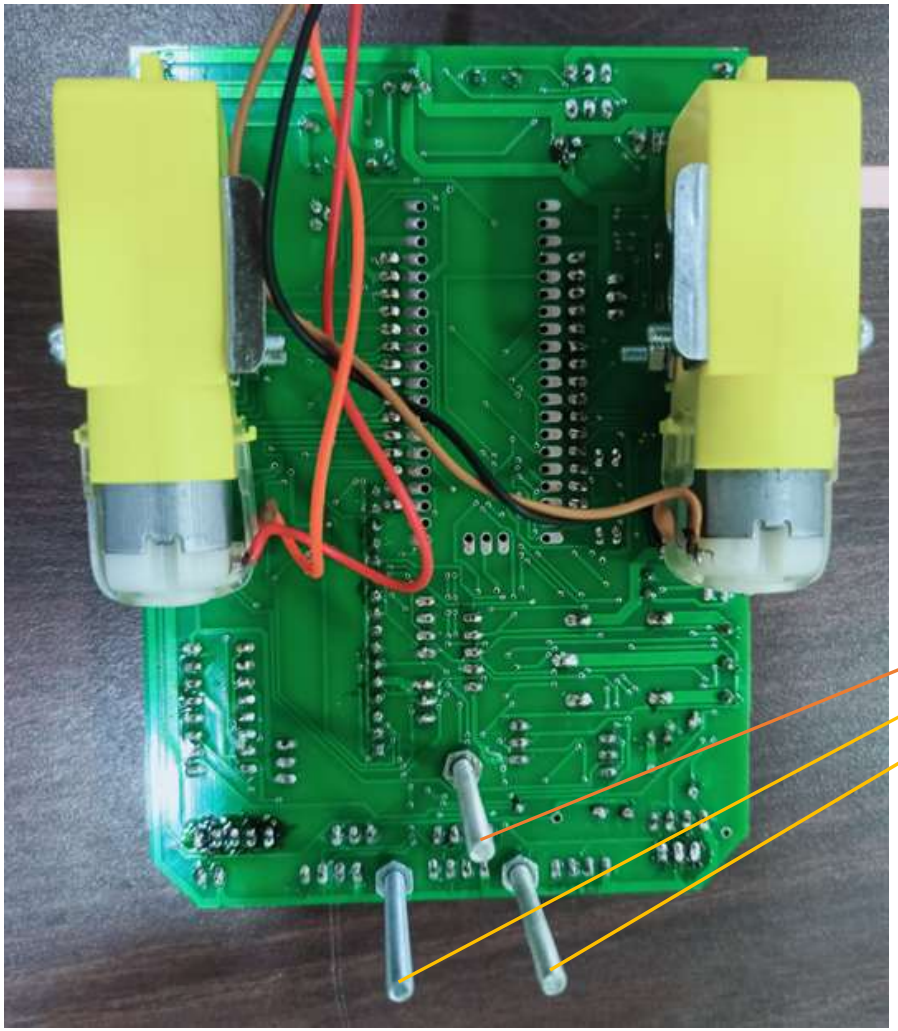
VJT303 MOTOR CLAMP & 10 mm Screws

Step-2:
Insert 25 mm Screws from the Bottom Side & Top Side of the 5V DC MOTOR (BO1 VJT389)





Step-3:
Fix the DC motor to the motor clamps using Nuts.



Step-4:
Fix the three 40 mm screws to the bottom side of the PCB using Nuts